Introduction to Statistics

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Statistics and Data Analysis

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What is Statistics?

- Statistics is the discipline concerned with the collection, organization, analysis, interpretation and presentation of data.
- Its objective is to extract useful information from data and support decision-making under uncertainty.
- Statistics can be divided into two major areas:
  - Descriptive statistics
  - Inferential statistics
- Statistical analysis provides a framework for understanding patterns, variability and relationships in data.

Descriptive and Inferential Statistics

- **Descriptive statistics** summarizes and describes the information contained in a dataset.

  Examples include:
  - Mean
  - Median
  - Standard deviation
  - Percentiles
  - Tables and graphs

- **Inferential statistics** uses information from a sample to make conclusions about a larger population.

- Inferential methods include estimation, hypothesis testing and regression analysis.

Population and Sample

- A **population** is the complete set of individuals, observations or units of interest.
- A **sample** is a subset of the population selected for analysis.
- Studying an entire population can be expensive, slow or impossible.
- Statistical inference allows us to use information from a sample to learn about the population.

Population $\rightarrow$ Sampling $\rightarrow$ Sample $\rightarrow$ Statistical inference

Parameter and Statistic

- A **parameter** is a numerical characteristic of a population.
- A **statistic** is a numerical characteristic calculated from a sample.

  Examples:
  - Population mean: μ
  - Sample mean: $\bar{x}$
  - Population variance: σ^2
  - Sample variance: s^2
- The statistic is often used to estimate the unknown population parameter.

Statistical Variables

- A **variable** is a characteristic that can take different values across observations.
- Variables can be classified as:
  - **Qualitative variables:** describe categories or attributes.
  - **Quantitative variables:** represent numerical quantities.
- Quantitative variables can be:
  - Discrete
  - Continuous

Types of Variables

- **Nominal:** categories without an intrinsic order.

  Example: country, gender, industry.

- **Ordinal:** categories with an ordered structure.

  Example: low, medium and high.

- **Discrete:** numerical variables based on counts.

  Example: number of firms.

- **Continuous:** numerical variables that can take values within an interval.

  Example: income, height or temperature.

Measures of Central Tendency

- Measures of central tendency describe the central or typical value of a dataset.
- The three most common measures are:
  - Mean
  - Median
  - Mode
- The appropriate measure depends on the distribution and characteristics of the data.

Arithmetic Mean

The arithmetic mean is calculated as:

$$\bar{x} = \frac{1}{n} \sum_{i=1}^{n} x_i$$

- x_i represents each observation.
- n is the number of observations.
- The mean uses every observation in the dataset.
- It is sensitive to extreme values.

For example:

$$2,\ 4,\ 6,\ 8,\ 10$$

$$\bar{x} = \frac{2+4+6+8+10}{5} = 6$$

Median and Mode

- The **median** is the value that separates the ordered data into two equal parts.
- The **mode** is the most frequently occurring value.

For the dataset:

$$2,\ 3,\ 3,\ 4,\ 5,\ 8,\ 10$$

$$\text{Mean} = 5$$

$$\text{Median} = 4$$

$$\text{Mode} = 3$$

The median is generally more robust than the mean when extreme observations are present.

Measures of Dispersion

- Measures of dispersion describe how spread out the observations are.

  Important measures include:
  - Range
  - Variance
  - Standard deviation
  - Interquartile range
- Two datasets can have the same mean but very different levels of variability.

Variance

The sample variance is defined as:

$$s^2 = \frac{1}{n-1}\sum_{i=1}^{n}(x_i - \bar{x})^2$$

- It measures the average squared deviation from the sample mean.
- Squaring the deviations prevents positive and negative differences from cancelling each other.
- The denominator $n-1$ provides the usual unbiased estimator of population variance under standard random-sampling assumptions.

Standard Deviation

The standard deviation is the square root of the variance:

$$s = \sqrt{\frac{1}{n-1}\sum_{i=1}^{n}(x_i - \bar{x})^2}$$

- Standard deviation is expressed in the same units as the original variable.
- A small standard deviation indicates observations are concentrated around the mean.
- A large standard deviation indicates greater dispersion.

Coefficient of Variation

The coefficient of variation measures relative dispersion:

$$CV = \frac{s}{\bar{x}}$$

or, expressed as a percentage,

$$CV = \frac{s}{\bar{x}} \times 100$$

- It is useful for comparing variability across variables with different scales.
- It is particularly useful when the mean is positive and meaningfully different from zero.

Frequency Distribution

- A frequency distribution describes how often different values or intervals occur.
- Frequencies can be represented using:
  - Frequency tables
  - Histograms
  - Bar charts
  - Density plots
- Visualization helps identify concentration, dispersion, skewness and unusual observations.

Normal Distribution

The normal distribution is one of the most important probability distributions in statistics.

Its probability density function is:

$$f(x) = \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{1}{2}\left(\frac{x-\mu}{\sigma}\right)^2}$$

- It is symmetric around the mean μ.
- The mean, median and mode are equal.
- The parameter σ determines the dispersion of the distribution.

The Empirical Rule

For a normally distributed variable:

- Approximately 68% of observations lie within $\mu \pm 1\sigma$.
- Approximately 95% lie within $\mu \pm 2\sigma$.
- Approximately 99.7% lie within $\mu \pm 3\sigma$.

$$68\% \quad \rightarrow \quad 1\sigma$$

$$95\% \quad \rightarrow \quad 2\sigma$$

$$99.7\% \quad \rightarrow \quad 3\sigma$$

Probability

- Probability quantifies uncertainty about the occurrence of an event.
- The probability of an event A satisfies:

$$0 \leq P(A) \leq 1$$

- A probability of 0 means the event is impossible.
- A probability of 1 means the event is certain.

Conditional Probability

Conditional probability measures the probability of an event given that another event has occurred.

$$P(A|B) = \frac{P(A \cap B)}{P(B)}$$

- $P(A|B)$ is the probability of A given B.
- $P(A \cap B)$ is the probability that both events occur.
- $P(B)$ is the probability of event B.

Correlation

- Correlation measures the strength and direction of a linear relationship between two quantitative variables.
- The Pearson correlation coefficient is:

$$r = \frac{\sum_{i=1}^{n}(x_i - \bar{x})(y_i - \bar{y})}{\sqrt{\sum_{i=1}^{n}(x_i - \bar{x})^2 \sum_{i=1}^{n}(y_i - \bar{y})^2}}$$

- The coefficient satisfies:

$$-1 \leq r \leq 1$$

Interpreting Correlation

- $r \approx 1$: strong positive linear association.
- $r \approx -1$: strong negative linear association.
- $r \approx 0$: weak or no linear association.

- Correlation does not by itself establish causality.
- A high correlation can arise from confounding variables, common trends or other mechanisms.

Sampling Methods

- Sampling determines which observations are selected from a population.

  Common probability sampling methods include:
  - Simple random sampling
  - Systematic sampling
  - Stratified sampling
  - Cluster sampling

- The sampling design affects the representativeness and precision of statistical estimates.

Sampling Error

- A sample generally does not reproduce the population perfectly.
- **Sampling error** is the random difference between a sample estimate and the corresponding population parameter.
- Increasing the sample size generally reduces sampling variability.
- However, increasing sample size does not automatically eliminate systematic bias.

Confidence Intervals

A confidence interval provides a range of plausible values for a population parameter.

For a population mean under standard conditions:

$$\bar{x} \pm z_{\alpha/2} \frac{\sigma}{\sqrt{n}}$$

- $\bar{x}$ is the sample mean.
- σ is the population standard deviation.
- n is the sample size.
- $z_{\alpha/2}$ is the appropriate critical value.

Hypothesis Testing

- Hypothesis testing provides a formal framework for evaluating claims about a population.
- The procedure usually involves:
  1. Formulating the null hypothesis H_0.
  2. Formulating the alternative hypothesis H_1.
  3. Choosing a significance level α.
  4. Calculating a test statistic.
  5. Obtaining a p-value or comparing with a critical value.
  6. Making a statistical decision.

Statistical Significance

- The p-value measures how incompatible the observed data are with the null hypothesis, under the assumptions of the statistical test.
- A common significance level is:

$$\alpha = 0.05$$

- If $p < \alpha$, the null hypothesis is rejected under the chosen testing rule.
- Statistical significance does not necessarily imply economic, practical or causal significance.

Simple Linear Regression

Simple linear regression models the conditional mean of a dependent variable as a function of one explanatory variable:

$$Y_i = \beta_0 + \beta_1 X_i + \varepsilon_i$$

- Y_i: dependent variable.
- X_i: explanatory variable.
- β_0: intercept.
- β_1: slope coefficient.
- ε_i: error term.

Interpreting the Regression Coefficient

For the model:

$$Y_i = \beta_0 + \beta_1 X_i + \varepsilon_i$$

- β_0 represents the expected value of Y when $X = 0$.
- β_1 represents the expected change in Y associated with a one-unit increase in X, conditional on the model specification.
- Interpretation depends on the measurement units, functional form and assumptions of the model.

Coefficient of Determination

The coefficient of determination is:

$$R^2 = 1 - \frac{SS_{Residual}}{SS_{Total}}$$

- R^2 measures the proportion of variation in the dependent variable explained by the fitted model relative to a constant-only benchmark.
- Higher R^2 does not necessarily imply a better causal model.
- Model quality should also consider assumptions, specification, prediction error and out-of-sample performance.

Key Statistical Concepts

- Statistics provides tools for understanding data and uncertainty.
- Descriptive statistics summarize observed data.
- Probability provides the mathematical framework for uncertainty.
- Sampling allows us to study populations using subsets of observations.
- Inferential statistics allows estimation and hypothesis testing.
- Correlation measures association, while regression provides a framework for modeling conditional relationships.
- Statistical results must always be interpreted considering assumptions, uncertainty and potential sources of bias.

Final Perspective

"Statistics is not only about calculating numbers."

It is about understanding variation,
quantifying uncertainty,
and making informed decisions from data.